SIMULTANEOUS WEAK GENERICITY FROM ESCAPING FUNCTIONS

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Abstract. A single total Turing functional Γ is constructed such that, for every c.e. set A and every A-escaping function f, the real Γ^f is weakly A-generic. Thus the same output works simultaneously for every c.e. oracle relative to which f is escaping.

Consequently, for every Turing ideal $\mathcal{I}$ with an internally cofinal family of c.e. sets, every $\mathcal{I}$-escaping function computes a weakly $\mathcal{I}$-generic real, without any assistance from the ideal.

In particular, an affirmative answer is given to a question raised by Bienvenu, Greenberg, Patey, and Turetsky immediately after the proof of their Proposition 7.1: the lowness assumptions on the generating c.e. chain can be removed. The uniformity requirement on that chain can also be eliminated.

The proof refines a distinguished-string construction of Andrews, Gerdes, and Miller by using an oracle-local bound on the lengths of competing strings, rather than deciding their eventual stability.

1. Introduction

A *Turing ideal* is a nonempty collection $\mathcal{I} \subseteq \omega^\omega$ closed downward under Turing reducibility and under finite Turing joins. Thus, if $X \in \mathcal{I}$ and $Y \leq_T X$, then $Y \in \mathcal{I}$; and if $X, Y \in \mathcal{I}$, then $X \oplus Y \in \mathcal{I}$. We identify subsets of ω with their characteristic functions. Let $\mathcal{I}$ be a Turing ideal, and call a function $f \in \omega^\omega$ *$\mathcal{I}$-escaping* if no total function in $\mathcal{I}$ eventually dominates f. A sequence $G \in 2^\omega$ is *weakly $\mathcal{I}$-generic* if it meets every dense set of binary strings c.e. relative to some element of $\mathcal{I}$. By $G \leq_{\mathcal{I}} f$ we mean $G \leq_T f \oplus Z$ for some $Z \in \mathcal{I}$.

Bienvenu, Greenberg, Patey, and Turetsky [3] ask which countable Turing ideals have the property that every $\mathcal{I}$-escaping function $\mathcal{I}$-computes a weakly $\mathcal{I}$-generic set.

Their Proposition 7.1 is an existence result: it asserts that there is a countable Turing ideal $\mathcal{I} \models \mathsf{HI}$ such that every $\mathcal{I}$-escaping function computes a weakly $\mathcal{I}$-generic set, without any assistance from $\mathcal{I}$.

Here $\mathcal{I} \models \mathsf{HI}$ means that $\mathcal{I}$ satisfies the *hyperimmune principle*: for every $X \in \mathcal{I}$, there is an X-escaping function belonging to $\mathcal{I}$.

The statement of that proposition does not involve lowness. However, its proof constructs the ideal from a strictly increasing sequence of uniformly c.e. and uniformly low sets. Immediately after the proof, the authors leave open whether the lowness requirements can be removed.

The question is answered affirmatively, and a stronger result is obtained. The argument does not rely on a low or uniformly presented generating chain. Instead, it establishes a universal conversion theorem which applies to every Turing ideal with an internally cofinal family of c.e. sets.

Date: November 8, 2026.

2020 *Mathematics Subject Classification.* Primary 03D30; Secondary 03D25.

Key words and phrases. Turing ideals, escaping functions, hyperimmunity, weak 1-genericity, computably enumerable oracles.

Theorem 1.1 (Universal simultaneous conversion)**.** *There is a single total Turing functional $\Gamma : \omega^\omega \to 2^\omega$ such that, for every c.e. set A and every total function f,*

$$f \text{ is } A\text{-escaping} \quad \Longrightarrow \quad \Gamma^f \text{ is weakly } A\text{-generic.}$$

The functional Γ does not receive an index for A.

Here f is *A-escaping* if $f \not\leq^* h$ for every total $h \leq_T A$, where $f \leq^* h$ means $f(n) \leq h(n)$ for all sufficiently large n. The order of quantifiers in Theorem 1.1 is important: for any fixed f, the *same* Γ^f is weakly A-generic simultaneously for all c.e. sets A relative to which f is escaping. We do not require f to escape functions computable from *all* c.e. sets.

Corollary 1.2 (C.e.-cofinal ideals)**.** *Suppose a Turing ideal $\mathcal{I}$ has an internally cofinal family of c.e. sets:*

$$(\forall X \in \mathcal{I})(\exists A \in \mathcal{I})\big[A \text{ is c.e. } \&\ X \leq_T A\big].$$

Then every $\mathcal{I}$-escaping function f computes a weakly $\mathcal{I}$-generic set. Indeed, Γ^f from Theorem 1.1 is weakly $\mathcal{I}$-generic. No effective presentation of the cofinal family is required.

Proof. If A is c.e. and belongs to $\mathcal{I}$, then every A-computable function belongs to $\mathcal{I}$. Hence an $\mathcal{I}$-escaping f is A-escaping, and Γ^f is weakly A-generic. For a dense set of strings D c.e. relative to $X \in \mathcal{I}$, choose a c.e. $A \in \mathcal{I}$ with $X \leq_T A$. The same D is A-c.e., so Γ^f meets D. □

Corollary 1.3 (Removing the lowness hypothesis)**.** *Let $(A_n)_{n\in\omega}$ be any sequence of c.e. sets with $A_n \leq_T A_{n+1}$, and set*

$$\mathcal{I} = \{X : (\exists n)\ X \leq_T A_n\}.$$

Every $\mathcal{I}$-escaping function computes a weakly $\mathcal{I}$-generic set. Neither lowness nor uniform enumerability of the sequence is needed; for this conclusion the degrees need not increase strictly.

Proof. The sequence $(A_n)_{n\in\omega}$ is an internally cofinal family of c.e. sets in $\mathcal{I}$. Thus the conclusion follows immediately from Corollary 1.2. □

Conversely, every Turing ideal with an internally cofinal family of c.e. sets is generated by some increasing sequence of c.e. sets: enumerate its c.e. members as $(C_n)_{n\in\omega}$ and take $A_n = C_0 \oplus \cdots \oplus C_n$. Thus the two corollaries describe the same class of ideals, although no uniform enumeration of the generating sequence is required.

The relationship between these results and Proposition 7.1 of [3] is worth emphasizing. The ideal constructed in the proof of that proposition has an internally cofinal family of c.e. sets, namely its defining chain. Corollary 1.2 isolates this structural property as a general sufficient condition, without requiring the cofinal family to be a chain or to have an effective presentation.

Corollary 1.3 then applies to the same form of ideal used in the original proof, but removes both the lowness and uniformity requirements on the generating sequence. Strict increase of the Turing degrees is unnecessary for the genericity conclusion; it is sufficient, however, to retain the additional property $\mathcal{I} \models \mathsf{HI}$. The construction is based on the distinguished-string and last-absence-cost method in the proof of Andrews–Gerdes–Miller [1, Theorem 3.3]. In their verification, the fixed oracle controls when competing candidates cease changing. Here we instead bound their *lengths*, allowing each dense requirement to be verified using only its own oracle. This "locality" is what allows us to include all c.e. oracles in a single effective list.

2. A local conversion lemma

We work in $2^{<\omega}$. For strings σ, τ, write $\sigma \preceq \tau$ if σ is an initial segment of τ, and $\sigma \prec \tau$ for proper initial segment. A set $U \subseteq 2^{<\omega}$ is *dense* if every string has an extension in U. A real G *meets* U if some $\sigma \prec G$ belongs to U.

The following lemma is the main technical ingredient of our proof. It separates the effective construction of the output from the oracle-dependent verification of individual requirements. Each requirement may have its own oracle, and neither the oracles nor their convergence moduli need to be given uniformly. Only the approximations to the sets of strings must be uniform. The crucial point is that a dense requirement can be satisfied using the escaping property relative to its own oracle, independently of the behavior of the other requirements.

Lemma 2.1 (Local conversion)**.** *Suppose $U_j \subseteq 2^{<\omega}$, $j \in \omega$, have a computable uniform approximation*

$$u(j,t,\sigma) \in \{0,1\}, \qquad \lim_t u(j,t,\sigma) = \mathbf{1}_{U_j}(\sigma).$$

For each j suppose an oracle B_j computes U_j and a total nondecreasing function m_j satisfying

$$t \geq m_j(L), \quad |\sigma| \leq L \quad \Longrightarrow \quad u(j,t,\sigma) = \mathbf{1}_{U_j}(\sigma). \tag{1}$$

Then there is a total Turing functional $\Psi : \omega^\omega \to 2^\omega$, depending only on u, such that for every j and f,

$$U_j \text{ is dense and } f \text{ is } B_j\text{-escaping} \quad \Longrightarrow \quad \Psi^f \text{ meets } U_j.$$

The reductions to B_j need not be uniform in j.

Proof. The construction. Given f, let

$$g(n) = n + 1 + \max_{a \leq n} f(a),$$

and put $g(-1) = -1$. Thus $g \leq_T f$, g is strictly increasing, and $g \geq f$ pointwise; every B_j-escaping assumption on f passes to g.

At the beginning of stage n we have an output string x_n of length n, together with zero or one *distinguished string* σ_j for each $j < n$. All these strings, including x_n, form a chain under extension. Initially x_0 is empty, and no strings are distinguished. Put $N = g(n)$ and perform the following steps.

(1) *Cancellation.* Remove each existing distinguished string σ_j for which

$$(\exists t)\big[g(n-1) < t \leq N \;\&\; u(j,t,\sigma_j) = 0\big].$$

Removal does not cancel strings assigned to other requirements.

(2) *Assignment.* While some index $j \leq n$ has no distinguished string, let J be the set of such indices and let τ be the longest string in the current chain (including x_n). Define

$$a_N(j,\sigma) = \max\big(\{0\} \cup \{t \leq N : u(j,t,\sigma) = 0\}\big),$$
$$c_N(j,\sigma) = \max\big\{|\sigma|, a_N(j,\sigma)\big\}.$$

Choose a pair (j,σ) of least cost among $j \in J$ and $\tau \prec \sigma$, breaking ties effectively; distinguish σ for j, and continue. No cancellations are performed inside this loop.

(3) *Output.* Let x_{n+1} be the prefix of length $n+1$ of the longest string in the resulting chain.

All steps are effective in g. To see that the minimization terminates, for any $j \in J$ the candidate $(j, \tau 0)$ has cost at most $C = \max\{|\tau| + 1, N\}$. A minimum can therefore be found by searching the finitely many strings of length at most C. Note that we permit strings that currently appear *outside* U_j; otherwise a nondense requirement could block the construction.

At most $n+1$ assignments occur, and the new requirement n must be assigned. Hence the final chain contains a proper extension of x_n, so step (3) is well-defined. The chain invariant is preserved, and the x_n are nested; set

$$G = \bigcup_n x_n = \Psi^f.$$

This is a total f-computable real. Any distinguished string that is never again cancelled is an initial segment of G.

A bound for one requirement. Fix k for which U_k is dense. The following calculations use B_k only, together with finite computable simulations of the construction.

For $M \in \omega$, use B_k to choose, for each of the finitely many strings τ of length at most M, an extension $\rho_\tau \succ \tau$ in U_k. Put

$$R = \max(\{M+1\} \cup \{|\rho_\tau| : |\tau| \le M\}), \qquad E_k(M) = \max\{R, m_k(R)\}.$$

Then $E_k(M) > M$; for each $|\tau| \le M$ there is $\rho \succ \tau$ such that

$$|\rho| \le E_k(M), \qquad (\forall t \ge E_k(M))\ u(k,t,\rho) = 1. \tag{2}$$

For $n \ge k$ and a strictly increasing positive history $v = (g(0), \dots, g(n-1))$ of length n, simulate the first n stages using v. Let L be the maximum of n and the lengths of the distinguished strings present just before stage n. Define

$$T_0 = L, \qquad T_{r+1} = E_k(T_r) \quad (0 \le r \le n),$$

and set

$$H_k(n, v) = 1 + \max(\{T_{n+1}, m_k(T_{n+1})\} \cup \{v(n-1) : n > 0\}). \tag{3}$$

This bound is B_k-computable from (n, v).

We claim that if $g \upharpoonright n = v$ and $g(n) \ge H_k(n, v)$, then by the end of stage n the requirement k has a true distinguished string which is never cancelled again.

Indeed, write $N = g(n)$. If an old σ_k survives cancellation, then $|\sigma_k| \le L$, $u(k, N, \sigma_k) = 1$, and $N \ge m_k(L)$. Its membership in U_k has stabilized, so it is true and will never be cancelled at a later stage.

Otherwise k is unassigned after cancellation. At the start of the assignment loop, the longest string has length at most T_0. Suppose inductively that just before the rth assignment, while k remains unassigned, the longest string has length at most T_{r-1}. By (2), requirement k offers a candidate extending this longest string with cost at most T_r: its length is at most T_r, and its last absence before time N occurred before T_r. Therefore the minimum-cost candidate chosen at the rth assignment has *length at most* T_r, irrespective of which requirement it belongs to or whether its eventual membership is correct. This gives the induction step. Notice that at each assignment substep the candidate for k is chosen anew above the *current* longest string: a candidate considered at an earlier substep may become incompatible when a different requirement is assigned. The repeated application of E_k absorbs this change, without requiring the candidates for any competing requirement to stabilize.

When k is assigned, its distinguished string σ has $c_N(k, \sigma) \le T_r$. If $\sigma \notin U_k$, then $N > m_k(T_{n+1}) \ge m_k(T_r)$ and $|\sigma| \le T_r$, so $u(k, N, \sigma) = 0$. Consequently $a_N(k, \sigma) = N > T_r$, a contradiction. Thus $\sigma \in U_k$, and its membership has already stabilized by time N. Since there are at most $n+1$ assignments, k is eventually assigned during the loop, proving the claim.

Removing the dependence on the history. The expression $H_k(n, g \upharpoonright n)$ need not itself be B_k-computable, so we cannot apply escaping to it directly. Fix the finite tuple $b = g \upharpoonright k$

as a numerical parameter. For this fixed b, recursively define a function $h_{k,b} \leq_T B_k$. Set $h_{k,b}(n) = 0$ for $n < k$. For $n \geq k$, let $V_n = V_{n,k,b}$ consist of the strictly increasing positive histories v of length n with

$$v \upharpoonright k = b, \qquad v(j) < h_{k,b}(j) \quad (k \leq j < n),$$

and set

$$h_{k,b}(n) = \max(\{1\} \cup \{H_k(n, v) : v \in V_n\}). \tag{4}$$

The search for V_n is finite: after the fixed first k coordinates, each coordinate lies in an interval bounded by a previously determined value of $h_{k,b}$. Each evaluation of H_k terminates because U_k is dense. Hence $h_{k,b}$ is total and B_k-computable for *every fixed numerical tuple* b; the index of this reduction may depend on b. We do not assume that B_k can recover b from g.

As g is B_k-escaping, it cannot be eventually dominated by the particular function $h_{k,b} \leq_T B_k$. Thus there is an $n \geq k$ with $g(n) \geq h_{k,b}(n)$. Choose the least such $n \geq k$. Then $g(j) < h_{k,b}(j)$ for all $k \leq j < n$, whence the actual history $g \upharpoonright n$ lies in V_n. By (4),

$$g(n) \geq h_{k,b}(n) \geq H_k(n, g \upharpoonright n).$$

By the preceding claim, requirement k has a permanent true distinguished string, hence G meets U_k. This argument only used the B_k-escaping property and did not require anything of the other oracles. As k was arbitrary, the lemma follows.

□

3. Simultaneously handling all c.e. oracles

We now apply Lemma 2.1 to obtain Theorem 1.1. The key step is to represent the effectively open sets relative to all c.e. oracles by a single uniformly computable approximation satisfying the hypotheses of the lemma. This is achieved by replacing each oracle-c.e. set of strings with a time-bounded representative generating the same open set and admitting a convergence modulus computable in the corresponding oracle.

Fix a standard effective enumeration $(A_i)_{i\in\omega}$ of *all* c.e. sets, with uniformly computable increasing finite approximations $(A_{i,t})_{t\in\omega}$. For any oracle A, let $(W_e^A)_{e\in\omega}$ enumerate the sets of binary strings c.e. relative to A, with a standard time-bounded oracle-machine enumeration.

For every pair (i, e) define the decidable-in-A_i set of strings

$$U_{i,e} = \left\{\sigma \in 2^{<\omega} : \begin{array}{l}\text{some } \tau \preceq \sigma \text{ is enumerated into } W_e^{A_i} \\ \text{within the first } |\sigma| \text{ computation steps}\end{array}\right\}. \tag{5}$$

The finite-time convention makes membership of a given σ decidable in A_i.

Lemma 3.1. *The open set generated by $U_{i,e}$ equals the open set generated by $W_e^{A_i}$. In particular, $W_e^{A_i}$ dense implies $U_{i,e}$ dense. Moreover, the $U_{i,e}$ have uniformly computable approximations whose pointwise convergence is bounded by a modulus computable in A_i, for each fixed (i, e).*

Proof. Every string in $U_{i,e}$ extends a string in $W_e^{A_i}$. Conversely, if an infinite sequence extends some $\tau \in W_e^{A_i}$, a sufficiently long initial segment has length greater than the stage at which τ was enumerated, and therefore belongs to $U_{i,e}$. The same extension argument proves density preservation.

Approximate $U_{i,e}$ by replacing A_i with $A_{i,t}$ in the time-bounded test (5). This gives a uniformly computable $u((i, e), t, \sigma)$.

To obtain an A_i-computable convergence modulus for strings of length at most L, simulate the true first L steps of the $W_e^{A_i}$ enumeration using oracle A_i. Let $q(L)$ be strictly larger than every oracle query made in those steps (put $q(L) = 0$ if there were no queries).

With A_i, determine the finite segment $A_i \upharpoonright q(L)$ and wait for a stage s_L at which every positive entry of that segment has appeared in the increasing c.e. approximation $A_{i,t}$. This search terminates and is computable in A_i.

For every $t \geq s_L$ we have $A_{i,t} \upharpoonright q(L) = A_i \upharpoonright q(L)$. By induction over the first L machine steps, the computation with $A_{i,t}$ then follows exactly the true computation: every oracle query is one of the true queries and receives the same answer.

Thus all time-bounded tests for $|\sigma| \leq L$ have stabilized. Taking $m_{i,e}(L) = \max_{\ell \leq L} s_\ell$ gives the required nondecreasing A_i-computable modulus. □

Proof of Theorem 1.1. Apply Lemma 2.1 to the sequence $(U_{i,e})_{i,e}$, enumerated via a fixed computable pairing, with $B_{(i,e)} = A_i$. Lemma 3.1 verifies its hypotheses. The resulting total functional Ψ is fixed independently of A; put $\Gamma = \Psi$.

Fix a c.e. $A = A_i$ and an A-escaping f. For every e for which W_e^A is dense, the corresponding $U_{i,e}$ is dense. The local lemma yields that Γ^f meets $U_{i,e}$, hence also W_e^A. Thus Γ^f is weakly A-generic. No assumption was made concerning the oracles attached to any other requirements. □

4. Consequences and limitations

Recall that $\mathcal{I} \models \mathsf{DNC}$ means that $\mathcal{I}$ satisfies the *diagonally noncomputable principle*: for every $X \in \mathcal{I}$, there is a diagonally non-X-computable function in $\mathcal{I}$.

Corollaries 1.2 and 1.3 follow from the theorem. When the c.e. chain in Corollary 1.3 has strictly increasing degrees, its ideal still satisfies the hyperimmune principle HI, independently of lowness: for every n the modulus of the c.e. enumeration of A_{n+1} is computable in A_{n+1} and is not eventually dominated by an A_n-computable function. Otherwise A_{n+1} would be computable from A_n.

More generally, any nonprincipal Turing ideal $\mathcal{I}$ with an internally cofinal family of c.e. sets satisfies HI.

Indeed, given $X \in \mathcal{I}$, first choose a c.e. $A \in \mathcal{I}$ with $X \leq_T A$. Nonprincipality gives $Y \in \mathcal{I}$ with $Y \not\leq_T A$, and cofinality applied to $A \oplus Y$ gives a c.e. $C \in \mathcal{I}$ with $A \oplus Y \leq_T C$. Thus $X \leq_T A <_T C$, and the enumeration modulus of C is an X-escaping function in $\mathcal{I}$.

Such an ideal fails DNC: an incomplete c.e. degree cannot compute an ordinary diagonally noncomputable function, by Arslanov's completeness criterion [2].

Moreover, if $\mathcal{I}$ contained a complete c.e. degree, then it would contain $\mathbf{0}'$. Since every c.e. set is computable in $\mathbf{0}'$, internal c.e. cofinality would imply that $\mathcal{I}$ is the principal ideal below $\mathbf{0}'$, contrary to nonprincipality.

Thus every c.e. set in $\mathcal{I}$ has incomplete Turing degree. By c.e. cofinality, no function in $\mathcal{I}$ can compute a diagonally noncomputable function, already relative to the computable oracle. Hence these ideals provide a broad class of examples in the region $\mathsf{HI} \wedge \neg\mathsf{DNC}$ considered in [3].

The construction can be relativized to an oracle Z. In particular, if $Z \in \mathcal{I}$ and $\mathcal{I}$ has an internally cofinal family of sets c.e. relative to Z, then every $\mathcal{I}$-escaping f has a weakly $\mathcal{I}$-generic $G \leq_T f \oplus Z$.

To see the oracle bookkeeping, an A_i which is c.e. in Z need not compute Z; the appropriate local oracle is $B_{(i,e)} = A_i \oplus Z$. This oracle belongs to $\mathcal{I}$, so f is $B_{(i,e)}$-escaping, and the

Z-relative version of the construction is computable from $f \oplus Z$. The conclusion does not assert $G \leq_T f$.

The argument does not characterize all countable Turing ideals $\mathcal{I}$ for which every $\mathcal{I}$-escaping function $\mathcal{I}$-computes a weakly $\mathcal{I}$-generic set, as asked in [3, Question 8.1].

Its key hypothesis is the availability of a convergence modulus *in the same oracle that decides each approximated dense set*. For arbitrary Δ^0_2 oracles, such moduli need not be available locally.

Acknowledgements

The main result was discovered by ChatGPT 6 Pro (OpenAI). The proofs were subsequently subjected to independent AI-assisted mathematical audits, including a detailed audit by Astra 6 Ultra (OpenAI). These audits were textual reviews, not formal verifications. The author takes full responsibility for the mathematical content.

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